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CADE: A lightweight change-adaptive dual-path encoder framework for predicting and generating future changes in satellite imagery using semi-supervised learning

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Abstract
Satellite image generation is critical in disaster management, environmental monitoring, and urban planning applications. Traditional methods of change detection often face challenges related to noise and atmospheric conditions, while recent deep learning methodologies, including Generative Adversarial Networks and transfer learning, typically require substantial computational resources. This study presents a novel, lightweight, Change-Adaptive Dual-Path Encoder (CADE) designed to mitigate these challenges, enabling the efficient generation of predictive satellite images with reduced resource demands. Our model enhances image generation capabilities by utilizing a Residual block in a separated dual encoder and attention module, even with limited labeled data. Additionally, semi-supervised training significantly strengthens the model's ability to learn from small datasets. CADE achieved a PSNR of 30.45 and an SSIM of 0.80 when tested on the LEVIR-CD dataset, surpassing top diffusion and GAN models. For DSFIN-CD, a PSNR of 28.93 and an SSIM of 0.76 were achieved. Furthermore, CADE obtained a 92.81% F1-score and 87.72% IoU on LEVIR-CD, and a 91.44% F1-score and 85.02% IoU on DSFIN-CD, all while using fewer parameters and reducing computational costs compared to other methods. This study contributes to Artificial Intelligence by proposing a semi-supervised encoder-decoder architecture that utilizes both labeled and unlabeled satellite imagery for temporal understanding and predictive modeling. Our findings indicate that CADE is precise and computationally efficient, showing that it is suitable for low-resource remote sensing uses. This semi-supervised method minimizes dependence on extensive annotated datasets as well. The use of this technology in engineering is pivotal for improving autonomous satellite systems that are responsible for effective Earth observation, evaluating natural disasters, and planning for sustainable environmental management.

Keywords Satellite · Dual-encoder · Generation · Semi-supervised

1 Introduction

The ability to accurately identify and predict changes on the Earth's surface using remote sensing has become increasingly essential for addressing global environmental and societal challenges. These insights, derived from satellite imagery, are important for informed decision-making across a wide range of applications, from monitoring agricultural productivity and managing natural disasters like floods, wildfires, and earthquakes to forecasting future environmental changes [1–4]. Furthermore, anticipating future landscape changes allows stakeholders to proactively manage risks such as land degradation. Although earlier approaches relied on basic imaging techniques such as differencing, graphical enhancement, and arithmetic-based transformations to monitor changes [5], these methods often fell short due to their sensitivity to atmospheric noise, sensor inconsistencies, and limited capacity for high-level feature extraction. Since the early 2010 s, computer-based information extraction has become a standard practice across a wide range of remote sensing applications [6]. However, detecting object-level changes and generating future projections of satellite imagery with higher precision remains a complex challenge, as low-level variations such as color shifts or rotational differences can obscure meaningful transformations in the scene [7].

Deep learning has become a transformative technology in the analysis of remote sensing data, addressing the drawbacks of traditional methods for tasks such as classification, detection, and segmentation [3, 4]. It has also enabled more accurate identification and localization of objects like buildings and vehicles in satellite and aerial images. One widely adopted strategy is transfer learning, which utilizes pre-trained models to extract relevant features from high-dimensional data and improve performance on remote sensing tasks [8]. Generative Adversarial Networks (GANs) have similarly gained attention for enhancing image generation and prediction in this domain [9]. These networks have been applied to augment datasets with synthetic imagery, simulate disaster scenarios, improve image resolution, and forecast future satellite observations [10, 11]. While GANs produce high-quality synthetic images, there is always a risk of artifacts or inaccuracies if the training data are limited [12]. In addition, unsupervised learning techniques have contributed to change detection by eliminating the need for labeled data and improving processing efficiency [13]. More recently, diffusion-based models have emerged as powerful tools for change detection and satellite image generation [14]. Denoising Diffusion Probabilistic Models (DDPMs), which try to capture meaningful representations from noisy data, have been particularly effective [15, 16]. Semantic Latent Diffusion Models have shown success in generating image samples for change detection [15]. Furthermore, transformer-based conditional generative diffusion models have combined the strengths of diffusion and transformer architectures to improve spatio-temporal modeling [17]. Hybrid architectures such as Siamese-Diffusion Networks have been employed to capture differences between pre- and post-event imagery [17–20], while Graph Representation Learning-Guided Diffusion Models have been introduced to leverage spatial and spectral structure for hyper-spectral change detection [21].

While existing approaches have significantly advanced the field, they frequently exhibit limitations that collectively restrict their broader applicability. For instance, transfer learning methods typically depend on domain similarity and still require considerable amounts of labeled data along with careful fine-tuning to perform effectively [8]. Similarly, GAN-based models often suffer from unstable training dynamics, incur high computational costs, and tend to perform poorly when data availability is limited [12]. Although unsupervised methods help reduce reliance on labeled data, they commonly struggle to extract meaningful change features from complex imagery. Likewise, diffusion models, despite their strong generative capabilities, generally require large datasets, extensive hyperparameter tuning, and longer training cycles, which make them computationally intensive and less scalable [15–17]. Moreover, hybrid architectures such as Siamese-Diffusion Networks and graph-based diffusion models introduce additional complexity [17, 21]. These challenges highlight the need for a solution that is not only accurate but also lightweight, scalable, and well-adapted to real-world, low-data environments. Moreover, most prior studies treat temporal change as a symmetric phenomenon, processing pre- and post-event images jointly or relying on dense bi-temporal supervision [4, 5]. This paradigm overlooks a critical asymmetry

inherent in real forecasting scenarios that future changes must be inferred from incomplete and uncertain change cues rather than observed outcomes [2]. As a result, existing models struggle to balance structural preservation with adaptive modification when predicting unseen future states [9–12].

While change detection has become a central task in remote sensing, it remains fundamentally retrospective, relying on the availability of post-event imagery to identify what has already occurred [2, 4, 5]. However, many real-world applications, such as disaster preparedness, urban growth management, infrastructure resilience planning, and environmental risk assessment, require anticipating future changes rather than reacting to past ones. In these contexts, the ability to generate plausible future satellite images provides a powerful means of visualizing potential scenarios before changes materialize, enabling proactive decision-making and early intervention.

Motivated by these gaps, this study reframes future satellite image generation as a change-adaptive reasoning problem, where static landscape memory and dynamic changes must be explicitly extracted and selectively fused. To address the challenge of generating accurate predictive satellite images with minimal computational demands, we propose a lightweight semi-supervised encoder-based model. Our approach uses a dual-encoder module to enhance feature extraction and representation. Additionally, we introduce a semi-supervised training approach, which significantly improves the model's ability to learn from small datasets, mitigating the need for large-scale labeled datasets that are often unavailable in remote sensing applications. The major contributions of our study are as follows:

- We propose a lightweight dual-path encoder architecture that separately processes static and dynamic regions in satellite imagery, allowing specialized feature extraction for both stable and change-prone areas.
- We introduce a novel cross-temporal attention fusion module that effectively aligns and merges the outputs of the static and dynamic encoders. This selective fusion enables the model to focus on regions likely to change, improving both the accuracy and perceptual quality of generated future images.
- Our approach leverages a semi-supervised training strategy to efficiently utilize both labeled and unlabeled data, making the model highly data-efficient and suitable for low-resource scenarios.
- Finally, our model shows robust generalization to out-of-distribution data, achieving high performance on both trained (LEVIR-CD) and unseen (DSFIN-CD) datasets without additional fine-tuning.

The rest of the paper is structured as follows: Section 2 examines prior research on change detection and predicting future scenes with satellite images. Section 3 covers the methodology that includes our proposed CADE architecture and training strategy. Section 4 elaborates on the experiments and result analysis. Section 5 discusses the strengths and limitations of this study. Lastly, Sect. 6 concludes the paper and suggests potential future research directions.

2 Related work

In this section, we review various deep learning approaches for remote sensing image processing, including GAN-based approaches, advanced encoder-decoder architectures, diffusion models, and attention-based change detection models.

2.1 Generative adversarial networks

Generative Adversarial Networks (GANs) have gained popularity in remote sensing data due to their powerful generative capabilities. For instance, Rui et al. [11] introduced DisasterGAN, which combines a disaster translation GAN and a damaged building generation GAN to synthesize diverse disaster scenarios from remote sensing data. Experimenting on the xBD dataset, they achieved Fréchet inception distance (FID) of 31.1684 and 21.7873 for the two networks. In another work, Zhao et al. [10] proposed a GAN-based encoder-decoder network for dehazing remote sensing images employing skip connections to merge high- and low-frequency generator

outputs and using a Markovian multi-scale discriminator to retain finer details. They achieved Peak Signal-to-Noise (PSNR) and Structural Similarity Index (SSIM) scores of 34.7389 and 0.9516 on the RICE-I dataset, and 32.9443 and 0.8515 on the RICE-II dataset, respectively. Gautam et al. [10] employed the Progressive Growing GAN (PGGAN) to produce high-resolution synthetic river images and addressed issues like a long training time and gradient problems seen in simpler GAN architectures. They achieved Laplacian Sliced Wasserstein Distance (SWD) scores between 0.00322 and 0.01527 across resolutions, with a 34.41% accuracy in distinguishing real from fake images. On the other hand, Liu et al. [22] proposed a Physics-informed Deep Adversarial Spectral Synthesis (PDASS) model that is based on the hyperspectral imaging model to predict subpixel-level abundance maps of ground objects. The framework achieved MSSIM and MPSNR scores of 0.9879 and 47.56 for the IEEE Grss_dfc_2018 dataset and MSSIM and MPSNR scores of 0.9956 and 60.21 for the GF5 dataset, respectively.

In a study, Jiang et al. [23] proposed CSIP-Net, an encoder-decoder-based CNN for predicting geosynchronous satellite observations. By integrating Residual Dense Blocks (RDB) and Convolutional Block Attention Modules (CBAM), the model captured spatio-temporal features and achieved 85.5% accuracy. Meanwhile, Wang et al. [12] proposed a GAN-based Conducted Semantic Embedding mode (CSEBGAN) for semantic-controllable remote sensing image synthesis that incorporates independent embeddings and a tripartite adversarial training scheme. They achieved an FID score of 155.29 and an accuracy of 94.10%.

Despite these innovations, GAN-based approaches often struggle with unstable training dynamics, mode collapse, and a heavy reliance on large volumes of labeled data. Moreover, the adversarial framework typically demands considerable computational resources and time to converge. Additionally, encoder-decoder architecture often employs deep and complex modules that increase computational load and training difficulty. Many such models rely on fully supervised learning and struggle to perform well when labeled data is sparse.

2.2 Diffusion-based models

More recently, diffusion models have emerged as popular alternatives for generative tasks in remote sensing. For instance, Xiao et al. [24] similarly designed a diffusion probabilistic model for high-quality remote sensing images using an Efficient Activation Network (EANet) with channel attention and gating operations for noise prediction. They also introduced a Conditional Prior Enhancement Module (CPEM) to extract informative cues from low-resolution inputs. On the DOTA dataset, the model achieved a PSNR of 28.30 and an SSIM of 0.7345. Zou et al. [25] proposed DiffCR, which combines conditional diffusion models and deep convolutional networks for cloud removal in optical satellite images. Their framework also introduced a Time and Condition Fusion Block (TCFBlock) to efficiently align input and target features.

Nevertheless, the strength of diffusion models often comes at the cost of efficiency. Their iterative denoising process leads to longer training times, increased sensitivity to hyperparameter tuning, and a greater need for large-scale training data.

2.3 Attention-based models

In earlier studies, attention modules have been integrated into CNN backbones. For instance, Liu et al. [26] designed DTCDSCN, which combines change detection and dual semantic segmentation networks with a dual attention module to enhance spatial and channel dependencies. The model reached an F1-score of 89.75% on the WHU building dataset. Similarly, Peng et al. [27] developed DDCNN, utilizing dense spatial-channel attention and a difference-enhancement unit to weight features adaptively. The DDCNN model achieved an F1-score of 90.24% on the LEVIR-CD dataset. Likewise, Zhang et al. [5] proposed the IFN model, which uses supervised attention fusion and a two-stream architecture for precise change map reconstruction. Addressing temporal consistency, Chen et al. [28] introduced STANet, applying spatial–temporal self-attention across multi-scale subregions to better capture object-scale variations and long-range dependencies. The model achieved an F1-score of 83.9% to 87.3% on the LEVIR-CD dataset. In another work, Fang et al. [29] introduced SNUNet-CD, which

employs a densely connected nested UNet with an ensemble channel attention module to improve deep supervision and fine edge detail.

Attention mechanisms and transformer-based architectures have recently gained popularity in change detection due to their ability to model long-range dependencies and enhance feature discrimination. For example, Chen et al. [30] introduced the BIT model, which compresses bi-temporal features into semantic tokens and models spatial-temporal context through a transformer encoder-decoder, achieving overall F1-scores ranging from 69.26% to 89.31% on change detection datasets. Similarly, Li et al. [31] presented TransUNetCD, a hybrid architecture combining CNNs with transformers, extracting global context from tokenized CNN features and restoring spatial precision with a difference-enhancing UNet decoder. The model scored average F1-scores of 91.11% and 66.62% on the LEVIR-CD and DSIFN-CD datasets, respectively. On the other hand, Zhao et al. [32] proposed SGSLN, an exchanging dual encoder-decoder framework that fuses bi-temporal features at the decision level, improving spatial localization and handling semantic inconsistencies in complex change scenarios. With an encoder feature size of 512 channels in the model, it achieved an F1-score of 92.33% on the LEVIR-CD dataset. In another study, Zhao et al. [33] proposed GeoFormer, which incorporates geometric priors into a Swin Transformer encoder. They also introduce a Laplace Attention Fusion module for robust bi-temporal feature merging and use a UNet++ decoder for detailed map generation. The model achieved average F1-scores of 91.72% and 91.09%, respectively, on the LEVIR-CD and DSIFN-CD datasets.

While attention-based models enhance feature representation and global context understanding, they often suffer from high computational costs, sensitivity to training data size, and challenges in preserving fine details, especially when changes are subtle or noisy. Moreover, most rely on fully supervised learning, limiting their scalability in low-label or cross-domain scenarios.

Existing literature on remote sensing primarily focuses on retrospective analysis, including change detection, segmentation, and image reconstruction [2, 4, 5]. While these approaches are valuable for understanding historical transformations, they do not address the equally critical task of forecasting future landscape evolution. Future satellite image generation extends beyond detecting changes to providing a visual and spatially coherent representation of how a region may evolve, which is essential for scenario-based analysis and decision support. However, this problem remains underexplored compared to traditional change detection tasks.

Although earlier research has advanced remote sensing image synthesis and change detection with GANs [9–11, 22, 23], diffusion models [24, 25], and attention-based networks [5, 26–29, 31–33]. These methods typically emphasize either image realism, segmentation performance, or dense bi-temporal supervision. Current generative approaches do not provide explicit mechanisms to disentangle invariant scene structure from evolving changes, and attention-based techniques are largely tailored to post-event change detection rather than forecasting future imagery. In addition, most existing methods depend on fully supervised training and are computationally intensive, restricting their use in resource-constrained or real-time settings.

To address these limitations, this work proposes a lightweight, change-adaptive dual-encoder architecture that explicitly decouples static landscape representations from dynamic change signals and integrates them via cross-temporal attention. By coupling semi-supervised learning with sparse changes, the proposed framework supports efficient, interpretable prediction of future satellite images without requiring post-change observations, thereby filling an important gap in the current literature.

3 Methodology

The proposed framework consists of three main components: (i) a dual-path encoder that processes the image and mask separately, (ii) a cross-temporal attention fusion module that aligns and merges the static and change features, and (iii) a decoder that reconstructs the future-change image. The model is trained using a masked reconstruction loss on unchanged regions and a consistency loss on unlabeled data, which enables it to learn from both labeled and unlabeled samples.

Figure 1 provides an overview of our proposed framework, specifically designed for future predictions based on satellite data.

3.1 Datasets

For this research, two datasets were used: the LEVIR-CD dataset and the DSFIN dataset. Both datasets are mainly change detection datasets and contain before and after images with corresponding change masks.

3.1.1 Data acquisition

LEVIR-CD. The LEVIR-CD [28] is a large-scale remote sensing Building Change Detection dataset. There are 637 very high-resolution (VHR, 0.5 m/pixel) images in this dataset. These bitemporal images, which cover a period of 5 to 14 years, show notable changes in land usage, particularly a rise in development. LEVIR-CD encompasses a wide range of building styles, including lofty flats, villa homes, little garages, and big warehouses. The fully annotated LEVIR-CD dataset contains a total of 31,333 individual change-building instances.

DSFIN-CD. The DSFIN-CD [5] dataset was gathered manually using Google Earth. It is composed of six sizable high-resolution bi-temporal images of six Chinese cities: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xian. The five major picture pairs (Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan) are each divided into 394 smaller image pairs. After this, 3940 bi-temporal pictures were obtained by augmentation, which was part of the dataset. For model testing, the picture pairs of Xian are divided into 48 image pairs. Therefore, the test set of this dataset is used for testing in order to deliver the model to unseen data. Table 1 overviews the datasets.

3.1.2 Data augmentation

Each image of the LEVIR-CD dataset was resized to 256×256 pixels to reduce memory requirements and time complexity during training. Our initial training dataset had only 637 images. As this is a very small number of images to train the model, data augmentation strategies were used to improve the training performance and robustness of our model. To guarantee consistency, the augmentations were applied to both the "before" and "after" satellite images and their masks. The augmentation process for the LEVIR-CD dataset is shown in Algorithm 1.

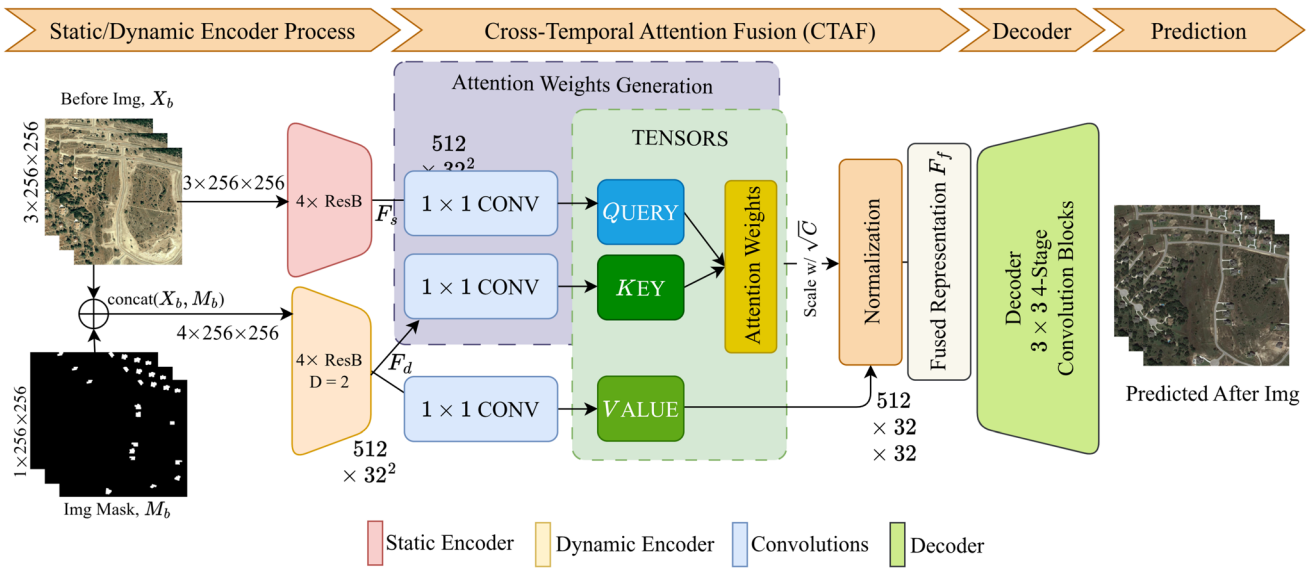


Fig. 1 Overview of the proposed methodology

Table 1 Comparison overview of the two datasets

Aspect	LEVIR-CD	DSFIN-CD
Data source	Google Earth	Google Earth
Resolution	0.5 m/pixel	High Resolution (Exact resolution not specified)
Image size	1024×1024 pixels (resized to 256×256)	512×512 pixels (resized to 256×256)
Geographical coverage	Varied, covering diverse areas over a period of 5 to 14 years	Six Chinese cities: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xian
Building types	Includes various building styles: tall flats, villa homes, garages, warehouses	General urban structures, no specific building types mentioned
Augmentation	✓	✗
After augmentation	822 images	✗
Training	All modified image + 30% of the masks	✗

Algorithm 1: Data augmentation strategies

Input: Original image pairs (X_b, X_a) and mask M (637 samples)
Hyperparameters: Flip $p = 0.5$; Saturation range: $[0.8, 1.2]$; Hue range: $[-0.1, 0.1]$; Iterations i

Initialize empty augmented dataset D_{aug}
For each original sample (X_b, X_a, M) in dataset:
  For $i = 1$ to 5:
    flip_horizontally $\leftarrow$ flip (X_b, X_a, M) with p
    flip_vertically $\leftarrow$ flip (X_b, X_a, M) with p
    Apply Color Jitter:
      brightness $\alpha \leftarrow$ uniform $(0.8, 1.2)$
      contrast $\beta \leftarrow$ uniform $(0.8, 1.2)$
      saturation $\gamma \leftarrow$ uniform $(0.8, 1.2)$
      hue $\delta \leftarrow$ uniform $(0.8, 1.2)$
    Adjust X_b, X_a using $(\alpha, \beta, \gamma, \delta)$
  Add augmented triplet (X_b, X_a, M_{aug}) to D_{aug}
Return D_{aug} (3822 samples)

This sequence was done five times, resulting in a training set of 3,822 images. The purpose of these modifications was to create a larger dataset and to replicate varying lighting situations and real-world scenarios, which would enhance the capacity of the model to adapt to diverse contexts [34].

3.2 Change-adaptive dual-path encoder

The proposed Change-Adaptive Dual-Path Encoder (CADE) is designed to extract complementary feature representations from the before-satellite image and the sparse change mask, without using the after-satellite image directly. CADE consists of two parallel encoding streams: the Static Encoder ϵ_s processes the before image X_b to extract spatial features that describe the unchanging or slowly changing environment; and the Dynamic Encoder ϵ_d encodes the sparse binary mask M_b and emphasizes areas where changes are likely or detected.

3.2.1 Static encoder

In satellite imagery, large portions of the landscape exhibit long-term consistency. To accurately depict a future image, a model must first understand and retain these stable patterns. In our framework, the Static Encoder is designed to analyze the input image independently of any explicit change signal. Its job is to encode the visual content of the pre-change image $X_b \in \mathbb{R}^{3\times H\times W}$ into a compact, multi-level representation that captures both low-level textures and high-level semantic structures.

The Static Encoder is composed of a series of Residual Blocks, introduced by He et al. [35], arranged hierarchically with progressively increasing feature dimensions: 64, 128, 256, and 512 channels. The encoder begins with a 3×3 convolution to process the RGB input and generate 64 feature maps. Each Residual Block is followed by a stride-2 convolution for downsampling, except for the first one, which retains the original resolution to preserve fine details. Batch Normalization and ReLU activations follow each convolutional operation to ensure nonlinearity and stable gradient flow. The final feature map carries deep semantic cues but remains spatially aligned with the input. These blocks are interleaved with strided convolutions to reduce spatial resolution while increasing representational capacity. This hierarchical structure allows the model to extract features at multiple levels of abstraction. Early layers capture fine-grained details such as edges, textures, and building outlines; intermediate layers learn mid-level features like shapes, layouts, and roads; and deeper layers abstract high-level semantics, such as land-use categories or urban–rural boundaries.

Each block within the encoder operates according to the Residual Block mechanism, ensuring that the input signal is preserved and refined rather than overwritten. The output tensor is defined as shown in Eq. (1):

$$F_s = E_s(X_b) \in \mathbb{R}^{512 \times H' \times W'} \tag{1}$$

where $H' = H/8$ and $W' = W/8$ represent the downsampled spatial resolution. In change detection tasks, not all information should be influenced by change signals. The Static Encoder acts as the memory of the model. It holds the blueprint of what the world looked like before the change. Later in the model, when features are fused with change-conditioned information, their representation helps anchor the network's imagination to real-world geography. Figure 2 depicts the static encoder block.

3.2.2 Dynamic encoder

While the Static Encoder teaches the model to remember the features as they are, it does not inform the model about what might be different in the future. Yet in many real-world cases, some rare changes happen, and we often have at least some indications of where to expect them. In our architecture, the Dynamic Encoder is designed to interpret contextual cues for change, using a combination of the pre-change satellite image and a corresponding binary change mask. The change mask $M \in \mathbb{R}^{1 \times H \times W}$ is a spatial hint that indicates where alterations may have occurred. It may come from various sources: automatic change detectors, auxiliary sensors, or even weak human annotations. We concatenate this binary mask with the original RGB input X_b to form a 4-channel input tensor. The process is defined in Eq. (2):

$$X_{bm} = \text{Concat}(X_b, M) \in \mathbb{R}^{4 \times H \times W} \tag{2}$$

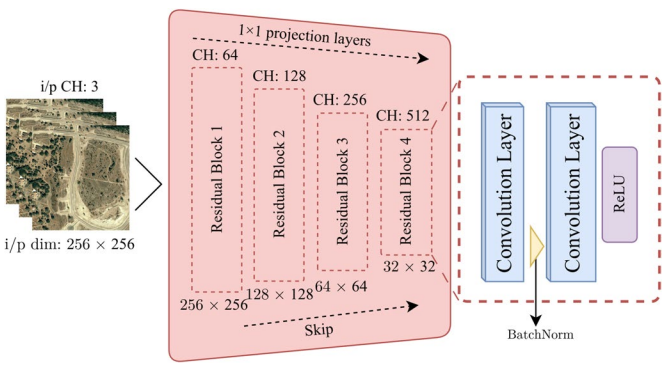


Fig. 2 The proposed static encoder block of our model. The dynamic encoder mirrors the static encoder's architecture and includes: a 4-channel (processes a 4-channel tensor formed by concatenating the RGB image and the binary mask) input image instead of 3; and the convolution is dilated with a dilation rate of 2

where M denotes the corresponding binary change mask, where a value of 1 indicates a potential change location, and 0 indicates unchanged regions. The operator $\mathrm{Concat}(\cdot)$ performs channel-wise concatenation and produces a four-channel tensor $X_{bm} \in \mathbb{R}^{4\times H\times W}$.

The Dynamic Encoder is guided to focus on spatial regions where modifications are likely to occur while still preserving the surrounding visual context. Structurally, it mirrors the Static Encoder in terms of depth and resolution but is specialized to learn change-aware representations. The encoder consists of an initial dilated convolutional Residual Block, which captures wide contextual information without reducing resolution. As with the Static Encoder, the feature map is progressively compressed to capture hierarchical semantics. We then process the concatenated input X_{bm} (from Eq. (2)) through a hierarchy of convolutional and residual layers to produce a deep feature representation. The output of the Dynamic Encoder F_d is a high-dimensional feature tensor, as defined in Eq. (3):

$$F_d = E_d(X_{bm}) \in \mathbb{R}^{512\times H'\times W'} \tag{3}$$

where $H' = H/8$ and $W' = W/8$. This downsampling factor (by 8) ensures that the dynamic features are spatially aligned with the output of the Static Encoder. Thus, it enables consistent feature interaction during the fusion stage. Some architectures fuse everything from the start. However, in our framework, we deliberately separate the static and dynamic pathways to decouple preservation from modification.

The Static Encoder is trained to preserve long-term structures. The Dynamic Encoder is trained to modulate that preservation when changes are detected or inferred. This division of labor improves specialization and helps the model make clearer decisions during the fusion process. Thus, the Dynamic Encoder is the model's change-sensitive interpreter, which analyzes the input image through the lens of possibility and prepares the necessary cues for downstream attention-based reasoning.

3.2.3 Cross-temporal attention fusion

The Static Encoder provides a holistic, memory-rich representation of the world as it currently exists. On the other hand, the Dynamic Encoder flags areas of possible transformation. Although these representations are powerful, they may carry conflicting information or emphasize different spatial patterns. If fused naively, they could also blur important details or even contradict each other.

To address this, we proposed the Cross-Temporal Attention Fusion (CTAF) module. CTAF is a learnable fusion mechanism that uses spatial attention to align and prioritize relevant features across both temporal paths. Rather than treating static and dynamic features equally across all regions, CTAF learns to dynamically focus on the most informative features. If a region is stable, it pays more attention to the static features. Likewise, if a region shows change signals, it shifts focus to the dynamic features. This selective reasoning is made possible through a dot-product attention mechanism, inspired by transformer architectures [36]. We first prepare both static (F_s) and dynamic feature F_d maps for attention-based fusion by reshaping them into flattened representations. As a result, the static feature (F_s) is mapped into F_s^{flat} and the dynamic feature into F_d^{flat}. Each spatial location is treated as a token, as shown in Eq. (4):

$$F_s^{\mathrm{flat}} = \mathrm{Reshape}(F_s) \in \mathbb{R}^{C\times N}, \quad F_d^{\mathrm{flat}} = \mathrm{Reshape}(F_d) \in \mathbb{R}^{C\times N} \tag{4}$$

where C is the number of channels, and N corresponds to the total number of spatial locations ($H \times W$). Following this, we compute the cross-temporal attention matrix, the attention map (A), using scaled dot-product attention, as defined in Eq. (5):

$$A = \mathrm{Softmax}\left(\frac{Q^T K}{\sqrt{d}}\right) \in \mathbb{R}^{N\times N} \tag{5}$$

where Q is the Query matrix formed from static features, and K is the Key matrix formed from dynamic features. The dot product $Q^T K$ measures the similarity between static and dynamic features across all spatial locations. To prevent extremely large dot-products (which stabilize gradients), the scaling factor d is included. Finally, the Softmax $(\cdot)$ function normalizes each row of the attention matrix such that the attention weights sum to one. Here, each static spatial location selectively attends to the most relevant dynamic locations.

This mechanism enables the model to emphasize change-sensitive regions while preserving static scene structure. We then computed the fused representation with the attention score A and the Value matrix V. The fused representation encodes a context-aware understanding of change and allows the decoder to create the future imagery. Through CTAF, each region in the static feature map sends a query to every region in the dynamic feature map, learns where to look for complementary or corrective information, and builds a context-enhanced, fused representation that reflects both long-term structure and short-term change. This process is entirely learnable and differentiable, and the attention weights are optimized during training via backpropagation.

3.2.4 Decoder and output generation

After the model has interpreted both what is stable (via the Static Encoder) and what is potentially changing (via the Dynamic Encoder), and reconciled the two perspectives through the Cross-Temporal Attention Fusion (CTAF) module, the Decoder translates this latent understanding into a full-resolution RGB image. The decoder reconstructs the future image $\hat{X}_e \in \mathbb{R}^{3\times H\times W}$ from the fused feature representation.

The input to the Decoder is the fused feature map $F_l \in \mathbb{R}^{C\times H\times H^N}$ with C number of channels, and spatial dimensions of H', W' ($H/8 \times W/8$, after encoding, see Eqd. (1) and (3)). The Decoder consists of a series of upsampling stages that double the spatial dimensions while reducing the number of channels. At each stage i, the input feature map $F^{[i-1]}$ is first upsampled using bilinear interpolation (doubling the spatial dimensions). A 3×3 convolution is then applied to refine spatial details and reduce aliasing artifacts, followed by a ReLU activation to introduce non-linearity. This process is repeated across multiple stages, which progressively restores the resolution from $H' \times W'$ back to the original image size $H \times W$ [37] Formally, the transformation is defined as presented in Eq. (6):

$$F^{(i)} = \mathrm{ReLU}\left(\mathrm{Conv}_{3\times 3}\left(\mathrm{Upsample}^{(F^{[i-1]})}\right)\right) \tag{6}$$

where $\mathrm{Upsample}(\cdot)$ is bilinear interpolation that doubles both spatial dimensions, $\mathrm{Conv}_{3\times 3}$ is a convolution that refines upsampled features. This structure enables the model to gradually reconstruct both coarse and fine spatial details. Once the spatial resolution has been restored, a 1×1 convolution projects the feature map into an RGB image. The process is shown in Eq. (7):

$$\hat{X}_a = \sigma\left(\mathrm{Conv}_{1\times 1}\left(F^{(3)}\right)\right) \in \mathbb{R}^{3\times H\times W} \tag{7}$$

where $\hat{X}_n$ denotes the predicted future satellite image, $F^{(3)}$ represents the output feature map of the final decoder stage, and $\mathrm{Conv}_{1\times 1}$ projects the decoder features from 64 channels to three channels corresponding to the RGB color space. The sigmoid activation function $\sigma(\cdot)$ ensures that the pixel intensities lie in the normalized range [0, 1]. This ensures the numerical stability during training and compatibility with real RGB satellite imagery. The decoder employs bilinear upsampling at each stage, and the subsequent 3×3 convolutions refine spatial details at each resolution level. It also enables the reconstruction of sharp boundaries and realistic textures. Residual learning in the earlier encoding stages ensures that high-quality structural features are preserved. This allows the decoder to focus primarily on refining fine-grained details rather than inventing scenes from scratch. A visual representation of our proposed decoder is shown in Fig. 3.

3.3 Semi-Supervised training

We used a semi-supervised approach [38] to train a model that will find potential changes and visualize them by generating new images based on the given images. For our dataset, we have a mask for each before-and-after image pair. The input of the model is the "before image," while the mask and the after image are the expected outputs. In this way, the model will be able to predict future changes and generate a visualization of these changes. 30% of the masks were used during the training process to help the model learn the specific feature patterns of the area changes. The remaining 70% of the masks were not used during training to ensure the prediction and generation capability of the model. Algorithm 2 represents the training loop.

Algorithm 2: Training algorithm of our semi-supervised model

Input: Model M, Labeled data (X_b, M_b, X_a), Unlabeled data X'_b
Output: Trained model M
Steps:

Initialize model M with random weights
for each epoch e from 1 to N **do**
  for each batch of labeled data (x_b, m_b, x_a) and unlabeled data x'_b **do**
    Forward pass (x_b) and (m_b) through the model: $\widehat{x_a} = M(x_b, m_b)$
    Compute supervised loss: $Loss = \text{MSE}(\widehat{x_a}, x_a)$
    for each unlabeled x'_b **do**
      apply mask m'_b with probability p: $m'_b \sim \text{Bernoulli}(p)$
    end for
    Forward pass x'_b and m'_b through the model: $\widehat{x'_a} = M(x'_b, m'_b)$
    Assume $\widehat{x'_a}$ is consistent with x'_b (no label available)
    Compute total loss for mini-batch: $\mathcal{C} = Loss$
    Update model M using gradient descent: $M \leftarrow M - \eta \nabla_M \mathcal{C}$
  end for
end for

4 Results

In this section, we provide a detailed analysis of the results for the two datasets and different configurations of our proposed model

4.1 Experimental setup

All experiments were conducted using the Kaggle cloud platform, which provides standardized, GPU-enabled environments. The following setup was used for training and evaluating the model and all baseline models. All models were trained and evaluated on a Kaggle Notebook instance equipped with an NVIDIA Tesla P100 GPU (16 GB VRAM), 2 Intel Xeon CPU cores, and 29 GB of RAM. The model was trained for 3000 epochs with a batch size of 8. The Adam optimizer was used with an initial learning rate of 1e-4, and a learning rate scheduler reduced the rate on a plateau. Early stopping was employed based on validation loss to prevent overfitting.

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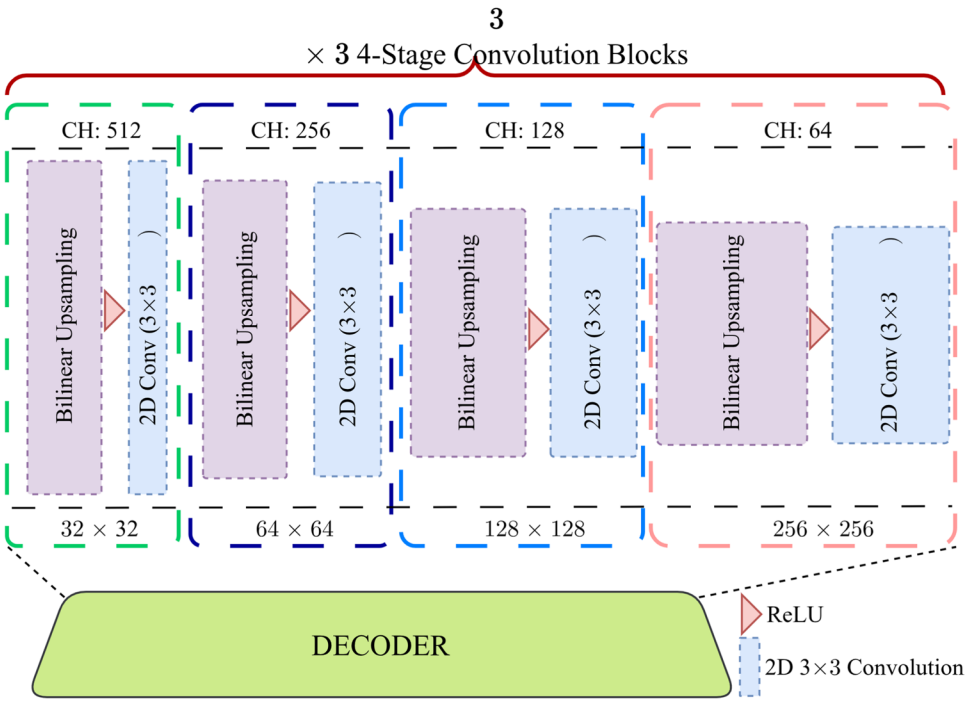


Fig. 3 Architecture of the 4-stage decoder block used in CADE. Each stage consists of bilinear upsampling followed by a 2D convolution (3×3) and ReLU activation. The feature maps are progressively upsampled from 32×32 to 256×256 spatial resolution, while the channel dimension decreases from 512 to 64. This design enables efficient recovery of spatial detail and refinement of image structure during the reconstruction of high-resolution satellite imagery

4.2 Evaluation metrics

To comprehensively assess the performance of our model, we initially evaluated it using standard segmentation metrics such as Precision, Recall, F1-score, and Intersection over Union (IoU). These metrics provide a foundational understanding of the model's accuracy, completeness, and overlap with ground truth segmentation masks.

Beyond these conventional metrics, we evaluated the quality of the reconstructed or generated images using perceptual and pixel-wise metrics, including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Fréchet Inception Distance (FID), and Learned Perceptual Image Patch Similarity (LPIPS). These metrics allow us to better understand both the quantitative fidelity and the perceptual quality of the images produced by our model.

Mean Squared Error (MSE). MSE is used to calculate the average squared difference between the actual values and their anticipated values by averaging the squared errors, where each error of each data point is the difference between its expected and actual values [39]. A lower MSE indicates that the predictions are closer to the actual values. Eq. (8) represents the Mean Squared Error:

$$\mathrm{MSE} = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}(I(i,j) - K(i,j))^2 \tag{8}$$

where I and K are the original and reconstructed images, respectively, with pixel values at position (i, j); and m and n are the dimensions of the images.

Peak Signal-to-Noise Ratio (PSNR). PSNR is the ratio of maximum potential signal power to noise, which degrades the image quality [39]. It is estimated by comparing an image to the best possible clean image with the highest power. The PSNR is defined as shown in Eq. (9):

$$\mathrm{PSNR} = 20 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I}{\sqrt{\mathrm{MSE}}}\right) \tag{9}$$

where MAX_I is the maximum possible pixel value of the image (e.g., 255 for an 8-bit grayscale image), and MSE is the mean squared error between the original and compressed image.

Structural Similarity Index (SSIM). The SSIM is used for measuring the similarity between two images. The SSIM index is a decimal value between −1 and 1, where 1 indicates perfect similarity. SSIM incorporates local patterns of pixel intensities that are strongly correlated with the quality of visual scenes by working on small windows or patches of an image [40]. Multiple windows of the image are used to calculate the index, and the distance between windows x and y is given by Eq. (10):

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{10}$$

where μ_x and u_y are the average values of x and y, σ_x and σ_y are the variances of x and y, σ_{xy} is the covariance of x and y, c_1 and c_2 are constants used to stabilize division with a small denominator.

Fréchet Inception Distance (FID). FID is a widely used metric for evaluating the quality and diversity of generated images [41]. It calculates the distance between the feature distributions of real and generated images in the embedding space, as shown in Eq. (11):

$$\mathrm{FID} = ||\mu_r - \mu_g||^2 + T_r\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right) \tag{11}$$

where μ_r and Σ_r are the mean and covariance of features from the real images, T_r is the trace of a matrix (i.e., the sum of its diagonal elements), and μ_g and Σ_g are the mean and covariance of features from the generated images. A lower FID score indicates that the generated images have similar statistics to real images, suggesting higher visual fidelity and diversity.

Learned Perceptual Image Patch Similarity (LPIPS). LPIPS is a perceptual similarity metric that compares images based on deep feature activations from neural networks. Unlike pixel-wise metrics, LPIPS captures semantic and perceptual differences more closely aligned with human visual judgment [42]. The LPIPS score is calculated using Eq. (12):

$$\mathrm{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h=1}^{H_l} \sum_{w=1}^{W_l} \|w_l \odot (\hat{f}_l^x(h, w) - \hat{f}_l^y(h, w))\|_2^2 \tag{12}$$

where $\hat{f}_l^x$ and $\hat{f}_l^y$ are the normalized feature activations from layer l for images x and y, w_l are learned channel-wise weights, and H_l and W_l are the height and width of the feature map at layer l. The operator $\odot$ denotes element-wise multiplication, while $\|\cdot\|_2^2$ represents the Squared Euclidean (L2) norm. A lower LPIPS value reflects higher perceptual similarity between the two images.

4.3 Training results

Table 2 reports the training outcomes for various model setups on the LEVIR-CD dataset. The experiments outline the progression of model performance with gradual structural improvements, assessed by Mean Squared Error (MSE), training loss, and validation loss. The initial base model (single encoder) resulted in an MSE of 0.083 and a validation loss of 0.51. Introducing Generative Adversarial Networks (GAN) [40] slightly enhanced the MSE to 0.049, while only minimally affecting validation loss. The dual-encoder model brought a notable decrease in MSE to 0.063, along with reduced training and validation losses. Incorporating pixel shuffle advanced the model's reconstruction ability, achieving an MSE of 0.019. Further integration of feature fusion and temporal attention mechanisms continued to improve performance, with temporal attention significantly decreasing the MSE to 0.009. In the end, the final model, utilizing a dual encoder with Cross-Temporal Attention Fusion (CTAF)

and a semi-supervised consistency loss, registered the lowest MSE of 0.001, indicating its superior predictive capability with training and validation losses at 0.11 and 0.14, respectively.

In our proposed model, the PSNR and SSIM values observed during training show a steady improvement in the model's performance. The PSNR gradually increases over the epochs and reaches a value of 30.45. Similarly, the SSIM approaches 0.80, suggesting that the generated images closely resemble the target images, with 1.0 representing perfect structural similarity. Figure 4 shows the PSNR and SSIM curves over 3000 (training) epochs.

4.4 Testing results

We conducted a thorough comparison of our model with a different state-of-the-art change detection algorithm on the LEVIR-CD dataset. Since we trained our model using a semi-supervised method, our model first intelligently predicts and generates a future mask. We evaluate our model based on four key metrics: Precision, Recall, F1-score, and Intersection over Union (IoU). The models chosen for comparison encompass a wide spectrum, including recent advancements such as transformer-based techniques (GeoFormer, TransUNetCD, SGSLN), as well as models incorporating advanced attention mechanisms (DTCDSCN, BIT, SNUNet-CD, DDCNN).

Each model offers distinct advantages, such as capturing long-range dependencies, multi-scale feature integration, or enhancing spatial and channel attention, collectively representing the forefront of automated change detection. Notably, our model consistently achieves superior or highly competitive results across all metrics. Specifically, it attains the highest Recall (93.76%), F1-score (92.81%), and IoU (87.72%) among the compared methods on the LEVIR-CD dataset. Its ability to outperform or closely rival leading current methods on a challenging standard underscores its architectural strength and relevance for practical remote sensing applications.

Table 3 indicates that CADE achieves competitive performance relative to recent change detection models that employ attention mechanisms and dual-path encoders. Methods such as GeoFormer, SGSLN, and report strong F1-score and IoU values by using explicit feature separation or attention-driven fusion. CADE follows an improved design direction (with cross-temporal attention fusion) yet reaches a competitive F1-score and IoU while using fewer parameters and lower computational cost (see Sect. 4.6.1). Additionally, its higher recall suggests improved sensitivity to change regions, while the balanced precision indicates that this improvement does not come at the cost of increased false positives.

We further assessed the generalization abilities of our model by analyzing its performance on the DSFIN-CD dataset. This dataset was only used for testing and not utilized for training or fine-tuning the model. Our model secures the highest scores across all four metrics, achieving a Precision of 93.01%, Recall of 89.92%, F1-score of 91.44%, and IoU of 85.02%. These enhancements over other top models are especially impressive given that our model was tested in a scenario without prior exposure to DSFIN-CD during training. The comparison in Table 4 shows that CADE preserves strong performance when evaluated on DSFIN-CD without dataset-specific training. Several existing models experience notable performance drops under this setting, particularly those trained in a fully supervised manner.

In contrast, CADE maintains a high F1-score and IoU, which indicates that the learned feature representations are not tightly coupled to the training data distribution. The performance of CADE on both LEVIR-CD and DSFIN-CD places it among the SOTA models for change detection. High F1-scores and IoU values, particularly

Table 2 Training results of different model configurations on the LEVIR-CD dataset

Model	MSE	Train loss	Val loss
Base Model (single Encoder)	0.083	0.44	0.51
+ Gen. Adversarial Networks	0.049	0.38	0.47
Dual-encoder	0.063	0.30	0.33
+ Pixel shuffle	0.019	0.34	0.38
+ Feature fusion	0.025	0.25	0.28
+ Temporal attention	0.009	0.17	0.21
Proposed Model	**0.001**	**0.11**	**0.14**

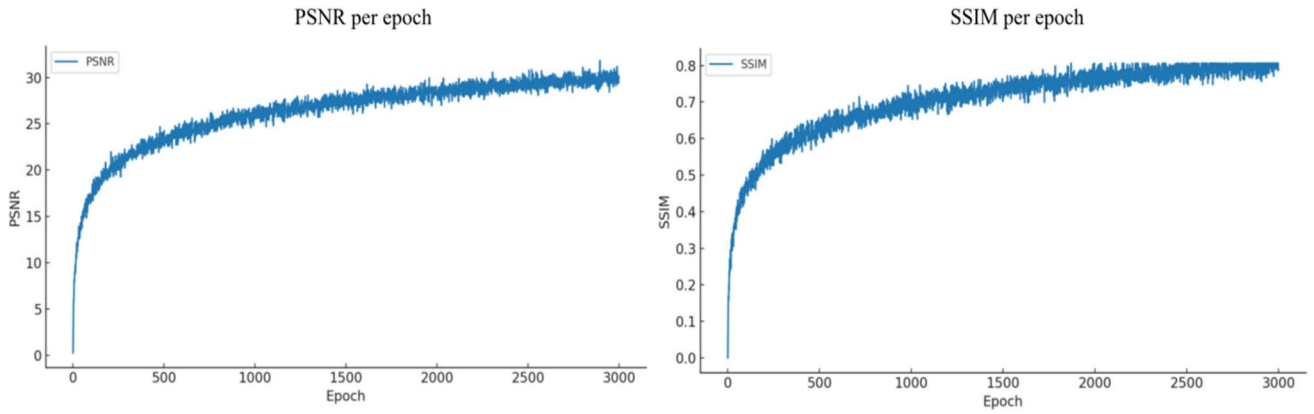


Fig. 4 PSNR and SSIM per epoch

Table 3 Comparison with state-of-the-art methods on the LEVIR-CD dataset

Methods	Precision	Recall	F1-score	IoU
GeoFormer [33]	92.83	80.17	83.40	71.53
DTCDSCN [26]	88.53	86.83	87.67	78.05
IFN [5]	**94.02**	82.93	88.13	78.77
STANet [28]	83.81	91.00	87.26	77.39
BIT [30]	89.24	89.37	89.31	80.68
SNUNet-CD [29]	89.18	87.17	88.16	78.83
TransUNetCD [31]	92.43	89.82	91.11	83.67
DDCNN [27]	91.85	88.69	90.24	82.21
FCCDN (512) [43]	93.07	91.52	92.29	85.69
SGSLN (512) [32]	93.07	91.61	92.33	85.76
Ours	91.89	**93.76**	**92.81**	**87.72**

The best and the second-best values are bolded and underlined, respectively. All results are in percentage (%)

on LEVIR-CD, are consistent with the outcomes of other leading dual-path and attention-based architectures. This also shows that specialized feature extraction and fusion mechanisms result in strong discriminative performance in satellite imagery analysis.

4.5 Future imagery generation

In this section, we evaluated our model's capacity to produce realistic, structurally precise future satellite imagery changes. Each primary architecture variation is assessed using four key metrics: SSIM, PSNR, FID, and LPIPS, which provide a comprehensive view of the generation quality of the model and its modules (Table 5). Transitioning from the basic single encoder to the fully advanced model offers a distinct and systematic enhancement in the model's generative capabilities. Figure 5 visualizes the generated future images.

Table 5 shows that initial setups are less effective at detailing fine features and subtle alterations, indicated by moderate SSIM and PSNR scores and relatively high FID and LPIPS values. With the adoption of a dual-encoder setup and feature fusion, the model shows improvement in extracting and integrating information necessary for predicting future changes, which causes a notable increase in SSIM and PSNR and significant reductions in FID and LPIPS. The incorporation of the Cross-Temporal Attention Fusion (CTAF) module boosts performance further.

As the model learns to focus on both static and dynamic regions strategically, it leads to more context-aware and visually coherent future forecasts. The final model version, featuring semi-supervised consistency loss, achieves the highest SSIM and PSNR and the lowest FID and LPIPS on both the LEVIR-CD and the DSFIN-CD out-of-distribution datasets. This result underscores the model's adeptness not only in grasping the dynamics of

Table 4 Comparison with state-of-the-art methods on the DSFIN-CD dataset

Methods	Precision	Recall	F1-score	IoU
DTCDSCN [26]	53.87	77.99	63.72	46.76
STANet [28]	67.71	61.68	64.56	47.66
IFN [5]	67.86	53.94	60.10	42.96
SNUNet-CD [29]	60.60	72.89	66.18	49.45
BIT [30]	68.36	70.18	69.26	52.97
TransUNetCD [31]	71.55	69.42	66.62	57.95
GeoFormer [33]	<u>91.41</u>	**90.78**	<u>91.09</u>	<u>83.64</u>
Ours	**93.01**	<u>89.92</u>	**91.44**	**85.02**

Our model was evaluated on the samples from this dataset (without specifically training on them). The best and the second-best values are bolded and <u>underlined</u>, respectively. All results are in percentage (%)

landscape change but also in generating high-fidelity and perceptually credible future satellite images, even in unfamiliar geographical areas.

4.6 Ablation study

4.6.1 Computation efficiency analysis

To analyze the model's complexity and inference efficiency, we conducted experiments with various configurations. This analysis is important for practical remote sensing applications, where computational resources, memory constraints, and real-time requirements often dictate the feasibility of large-scale deployment. Table 6 presents a detailed report on the model's parameters and inference time.

We reported three key indicators: (i) the number of parameters (in millions), (ii) the computational cost measured by floating point operations (FLOPs, in billions), and (ii) the average inference time per image (in seconds). The comparative models UNet++, BIT, FCCDN, and SGSLN are prominent in the literature for their performance in change detection tasks, yet they typically demand significant computational overhead. The results show that CADE, particularly in its shared encoder and dual encoder configurations, achieves notable reductions in both parameter count and FLOPs compared to these benchmarks. Even the final model incorporating the Cross-Temporal Attention Fusion (CTAF) module maintains a favorable balance between model size (5.93M parameters) and computational demand (10.89G FLOPs), with an average inference time of just 134 seconds, which is also consistently faster than most competing methods. By keeping the model lightweight and reducing inference time, the proposed model is well-positioned for real-world remote sensing scenarios, including operational monitoring and rapid response applications, where both high performance and efficiency are paramount.

4.6.2 Model's configurations and hyperparameter analysis

We also conducted a comprehensive ablation study to assess the influence of each key architectural and training component within the proposed model. The experimental configuration setup examines a variety of experimental setups by modifying the encoder configuration (single vs. dual encoder, with or without dilation), fusion methods (none, concatenation, additive fusion, or Cross-Temporal Attention Fusion (CTAF), optimizer options (Adam vs. SGD), and learning rate parameters.

The findings reveal several critical trends. For instance, shifting from a single to a dual encoder configuration significantly reduces loss, underscoring the importance of independently modeling static and dynamic elements. Enhancing the model with dilated convolutions and experimenting with fusion strategies, especially as seen in the shift from basic concatenation or addition to CTAF. The lowest losses are consistently observed in configurations featuring the dual encoder with CTAF fusion and the Adam optimizer, designating this arrangement as the most effective for modeling intricate change dynamics. The ablation findings validate the integration of the dual

Table 5 Image quality evaluation of the model's variants on the LEVIR-CD and DSFIN-CD datasets

Model variant	LEVIR-CD dataset				DSFIN-CD dataset			
	SSIM↑	PSNR↑(dB)	FID↓	LPIPS↓	SSIM↑	PSNR↑(dB)	FID↓	LPIPS↓
Baseline (single encoder)	0.602 ± 0.145	23.71 ± 0.34	68.52 ± 1.91	0.276 ± 0.017	0.578 ± 0.013	22.93 ± 0.41	74.85 ± 2.14	0.289 ± 0.019
+ Dual-Encoder (Static & Dynamic)	0.681 ± 0.024	24.86 ± 0.30	56.78 ± 1.67	0.243 ± 0.013	0.667 ± 0.011	24.08 ± 0.38	61.70 ± 1.85	0.254 ± 0.021
+ Feature Fusion w/Concatenation	0.709 ± 0.049	26.12 ± 0.28	48.04 ± 1.54	0.212 ± 0.021	0.691 ± 0.010	25.31 ± 0.36	52.13 ± 1.76	0.223 ± 0.018
+ Cross-Temporal Attention Fusion (CTAF)	0.736 ± 0.018	27.64 ± 0.26	37.93 ± 1.29	0.179 ± 0.010	0.718 ± 0.009	26.85 ± 0.33	41.78 ± 1.49	0.192 ± 0.015
+ Semi-Supervised Consistency Loss (final)	0.755 ± 0.011	28.78 ± 0.23	31.61 ± 1.13	0.162 ± 0.008	0.743 ± 0.008	28.04 ± 0.29	34.45 ± 1.37	0.173 ± 0.011

All metrics are reported as mean ± standard deviation on three runs. The results show consistent improvements in both fidelity (PSNR, SSIM) and perceptual quality (FID, LPIPS) as architectural and training components are introduced

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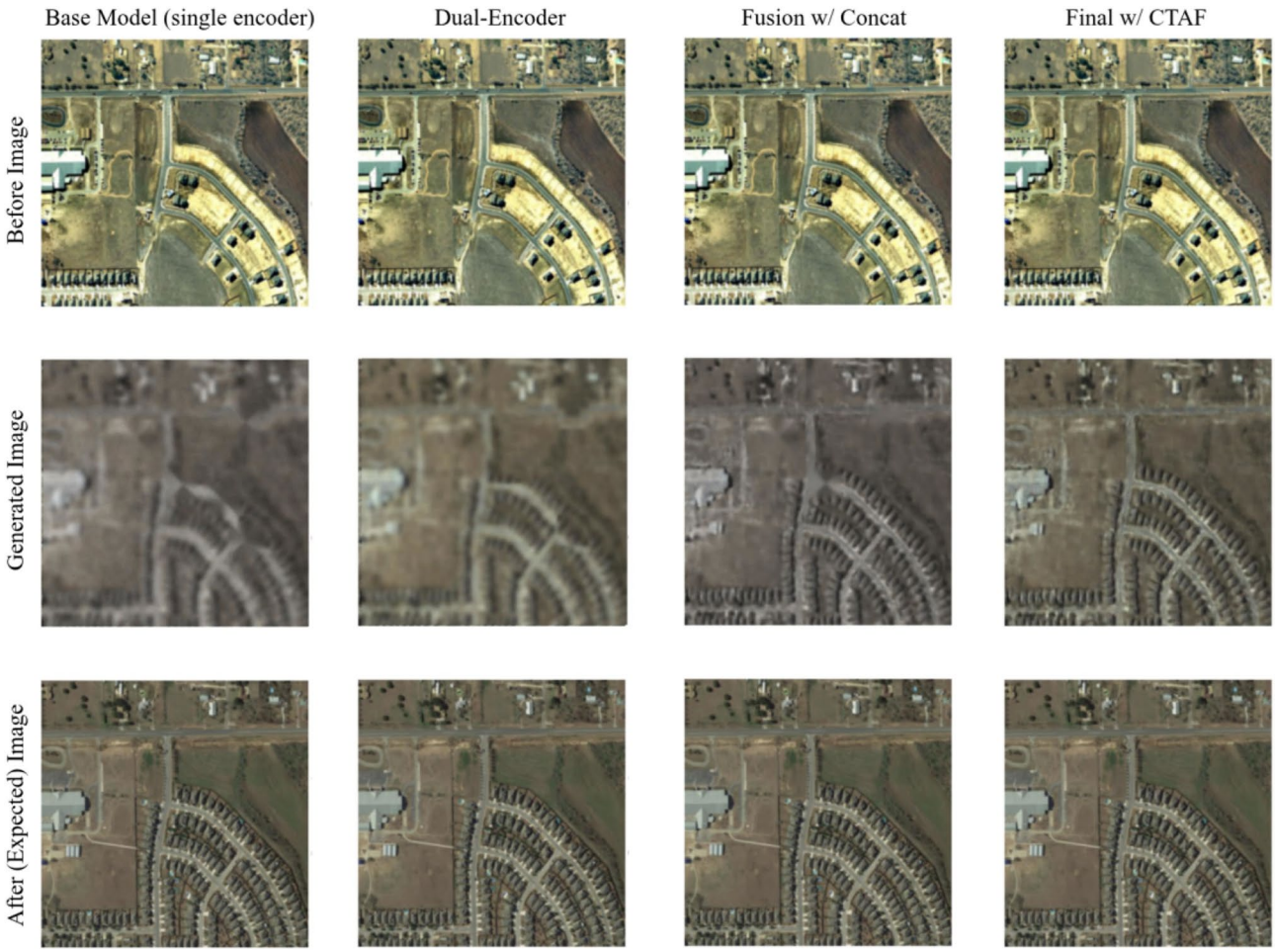


Fig. 5 Visualization of output from different experimental models

Table 6 Model complexity and inference time comparison

Model	Params (M)↓	FLOPs (G)↓	Inf. Time (s)↓
UNet++ [44]	9.05	34.01	170
BIT [30]	3.55	67.80	145
FCCDN (512) [43]	6.25	12.40	150
SGSLN (512) [32]	6.04	11.50	165
Ours: Shared Encoder	2.84	8.88	117
Ours: Dual Encoder	4.78	10.01	129
Ours: w/CTAF (final)	5.93	10.89	134

encoder and attention-based fusion into the final model and provide valuable insights for future endeavors seeking to optimize complexity and performance balance in change detection networks. The detailed model configurations and associated results are presented in Table 7.

Table 7 Ablation of the major components' configuration

Exp	Encoder setup	Fusion method	Optimizer	LR	Train loss↓	Val loss↓
1	Single encoder (shared)	✗	Adam	1e-4	0.243	0.281
2	Dual encoder (w/o dilation)	✗	SGD	1e-3	0.198	0.233
3	Dual encoder (dilated mask)	✗	Adam	1e-4	0.171	0.214
4	Dual encoder	Concatenation	Adam	1e-4	0.161	0.201
5	Dual encoder	Additive	Adam	1e-4	0.155	0.194
6	Dual encoder	CTAF	SGD	1e-3	0.138	0.167
7	Dual encoder	CTAF	Adam	1e-4	**0.110**	**0.141**
8	Dual encoder	CTAF	SGD	1e-3	0.148	0.177
9	Dual encoder	CTAF	Adam	1e-4	0.117	0.148
10	Dual encoder	CTAF	Adam	1e-4	0.114	0.143

The ablation experiment was conducted on the LEVIR-CD dataset. The best configuration is highlighted

5 Discussion

In this study, we propose a novel semi-supervised, lightweight encoder-based framework for predicting and generating future changes in satellite imagery. Unlike conventional remote sensing methods that focus on retrospective change detection, the proposed approach is designed to forecast plausible future landscape evolution under limited supervision and computational constraints. This formulation addresses the growing need for proactive analysis in applications such as disaster preparedness, urban planning, and environmental monitoring, where timely insight is essential.

The proposed framework enables proactive remote sensing analysis by generating future satellite images rather than relying on retrospective change detection. This capability has several practical applications. In disaster preparedness and risk assessment, future image generation can support early identification of regions likely to experience structural or environmental changes, aiding contingency planning and resource allocation. On the other hand, in urban planning and infrastructure management, the model can be utilized to simulate urban expansion and land-use transitions, supporting long-term development strategies. Finally, in environmental monitoring, the approach facilitates forecasting of gradual landscape changes such as deforestation, shoreline variation, or habitat degradation.

The novelty of the proposed framework lies in a change-adaptive dual-encoder architecture that explicitly disentangles static scene representations from dynamic change cues. Our primary innovation lies in the incorporation of a lightweight encoder model that integrates a static and dynamic encoder block into the architecture. This separation enables the model to avoid the common pitfalls of over-smoothing or feature dilution, which can occur when all information is fused prematurely. The following Cross-Temporal Attention Fusion (CTAF) mechanism facilitates the context-aware amalgamation of these feature streams, enabling the decoder to concentrate on areas most prone to change, while maintaining the integrity of unchanged regions.

A key strength of CADE is its lightweight design, which makes it highly suitable for real-world deployment where computational resources are limited. The final model, incorporating the Cross-Temporal Attention Fusion (CTAF) module, contains only 5.93 million parameters and requires 10.89 GFLOPs per inference, with an average inference time of just 134 seconds per image. This proves the practicality of CADE for time-sensitive or resource-constrained remote sensing applications. Additionally, our study introduces a novel semi-supervised training strategy that enhances model effectiveness by ensuring that the learned representations are highly informative about the target labels. This semi-supervised training method provides a more effective learning process, especially in scenarios with limited labeled data.

Compared with other SOTA models, our results also indicate a substantial advance in change detection. On the LEVIR-CD dataset, the proposed model achieves an F1-score of 92.81% and an IoU of 87.72%, which are higher than those of GeoFormer (83.40% F1, 71.53% IoU) and SGSLN (92.33% F1, 85.76% IoU). On the DSFIN-CD dataset, which was not used for training, the model attains an F1-score of 91.44% and an IoU of 85.02%. These results show strong cross-dataset generalization and highlight the model's ability to handle unseen data effectively. At the same time, the approach maintains efficiency, requiring only 5.93 million parameters and 10.89 GFLOPs, compared with UNet++ (9.05M, 34.01 GFLOPs) and BIT (3.55M, 67.80 GFLOPs), and it processes each image in 134 seconds. This combination of high accuracy, reliable generalization, and low computational cost makes the method well-suited for large-scale and resource-constrained remote sensing applications.

In addition to this, CADE delivers consistently impressive outcomes across other quantitative metrics. On the LEVIR-CD dataset, the model achieves a PSNR of 30.45 and an SSIM of 0.80 for image generation. Likewise, on the DSFIN-CD dataset, CADE achieves a PSNR of 28.93 and an SSIM of 0.76, illustrating strong generalization to out-of-distribution data. For perceptual image quality, our method results in strong Fréchet Inception Distance (FID) and Learned Perceptual Image Patch Similarity (LPIPS) scores. On LEVIR-CD, CADE achieves an FID of 31.61 and an LPIPS of 0.162; on DSFIN-CD, FID is 34.45, and LPIPS is 0.173. These values indicate that the generated future images are not only quantitatively accurate but also visually coherent and realistic.

Nevertheless, some limitations persist. The model's performance may still be constrained by the inherent quality and representativeness of available training data, and the use of binary masks as change priors may not capture all relevant dynamics in highly complex or ambiguous scenes. Future work could explore the integration of multi-modal data (e.g., SAR, multispectral) and more advanced attention or memory modules to further enhance temporal reasoning and adaptability.

Future work can focus on further enhancing the model's adaptability by integrating multi-modal and multi-temporal data and exploring more advanced attention mechanisms for complex scene understanding.

6 Conclusion

In this study, we introduced a novel lightweight encoder-based model. By leveraging a residual-block-based dual-path encoder, cross-temporal attention fusion, and a semi-supervised training paradigm. The proposed model effectively balances accuracy, computational efficiency, and data efficiency. Our proposed model not only provides effective image generation and future satellite imagery predictions but also minimizes the need for high computational resources commonly required by other existing approaches. On the LEVIR-CD dataset, CADE achieved an F1-score of 92.81%, IoU of 87.72%, PSNR of 30.45, and SSIM of 0.80, while on the DSFIN-CD dataset, it attained an F1-score of 91.44%, IoU of 85.02%, PSNR of 28.93, and SSIM of 0.76. At the same time, its lightweight architecture with under 6 million parameters and low computational demands ensures fast and efficient inference. Extensive experimental evaluation shows that the proposed approach achieves competitive performance while requiring lower model complexity and reduced reliance on labeled data compared to existing studies. Importantly, CADE's semi-supervised approach allowed it to learn effectively from limited labeled data, delivering strong generalization and reliable future image generation even on unseen data. By providing a more accurate and efficient method for monitoring changes in satellite images and generating reliable images for predicting future changes, our research supports informed decision-making in the field of remote sensing image analysis.

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Data availability In this study, two public datasets, LEVIR-CD [28] and DSFIN-CD [5], are utilized, available in Google Drive and Open Data Lab, respectively.

Declarations

Informed consents All the authors have read the manuscript fully and contributed to this research. The authors also provide consent for peer review and publication of this study.

Competing interests On behalf of all authors, the corresponding author states that there is no conflict of interest.

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